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CS-499-13445-M01 Computer Science Capstone

October 18, 2026

Professional Self-Assessment

This document is the front of my CS-499 ePortfolio. My concentration is software engineering. The three artifacts behind it are CS-320 Contact (software design and engineering), CS-300 ABCU catalog (algorithms and data structures), and CS-340 AnimalShelter for Grazioso Salvare (databases). The milestone narratives are already written. This page is the introduction to those artifacts. The live site is https://amaurihopewell.github.io/CS-499/. Original files stay on main. Each enhancement stays on its milestone branch.

Coursework, strengths, and employability

The capstone made me put three old projects in front of a reader who was not in the original class. CS-320 taught me to keep a field rule and test it. CS-300 taught me to time a lookup before I pick a structure. CS-340 taught me to get a password out of a constructor. The ePortfolio is where those three have to sit together and still be readable.

I am aiming at software engineering internships and junior or SDET roles in Washington, D.C. and Northern Virginia, on-site, hybrid, or remote, before graduation and for May 2027. The postings I read for the 4-1 journal asked for Java or Python, Git, SQL, and tests, and they expected an HTTP status code. ContactApi, the CS-300 compare run, and the AnimalShelter pytest suite are the files I can point to.

Amazon returns in Lexington is the example from outside these artifacts. That job taught me to look at a package before the next person opens it. I use the same check on an API body, on a delete that can wipe a collection, and on a README that tells an instructor how to run the code.

Collaborating in a team environment

The Module Two code review is a video of the original CS-320 Contact service: the field limits, the ArrayList next to a README that said HashMap, and the println in getContact. I recorded it so a reviewer can see the code without cloning first. After that I kept the original files on main and put each enhancement on its own branch: cs499-milestone-two, cs499-milestone-three, and cs499-milestone-four. A reviewer can diff them. I asked for comments on the Contact auth filter (401 versus 403, empty body), on the CS-300 median-first BST and menu 6, and on the AnimalShelter unique index when the AAC dump already has duplicate ids. The grades came back without a request for more code. The live site links main and those branches. The branches are not merged into main, so main is still the original upload.

Communicating with stakeholders

Grazioso staff need a filter they can open. dashboard.py is that screen: Water, Mountain, and Disaster rescue, a table, and a breed pie. The default role is staff, which can read. Admin is required to write. For a hiring manager the same idea is the README at the top of each enhanced folder: how to install, how to test, and how to open the UI. The milestone narratives are the longer version, written for the instructor and for someone reading the portfolio later. The 3-2 narrative is CS-499_3-2_Narrative_Software_Design_Amauri_Hopewell. The 4-2 narrative is CS-499_4-2_Narrative_Algorithms_Amauri_Hopewell. The 5-2 narrative is CS-499_5-2_Narrative_Databases_Amauri_Hopewell.

Data structures and algorithms

CS-300 is the ABCU course catalog. The original program kept courses in a vector. The enhancement times that vector against a BST and a hash map on the 129-course file, and it prints a topological schedule (menu 6) from the prerequisite list. A sorted insert into a BST becomes a chain. I fill the tree from the median of the sorted ids so the height stays logarithmic on that file. Compare mode only means something if the binary is built with /O2. The enhanced README says that.

Software engineering and databases

CS-320 ContactService still owns the field rules from the original class. The id is final. Names are at most 10 characters. The address is at most 30. The phone is ten digits. I put a repository under that service. Tests use an in-memory HashMap. The running service uses SQLite, so a contact is still there after a restart. ContactApi is a JDK HttpServer. I did not use Spring, so the HTTP layer is one class. mvn test is 37 tests.

CS-340 AnimalShelter is the database artifact. create, read, update, and delete go through one module. animal_id has a unique index. Lookups go through an allow-list, and an operator key such as $where is rejected. Staff and admin share one Mongo login from the environment. The dashboard role decides who can write. python -m pytest tests -q is 12 tests on mongomock. If MONGO_USER and MONGO_PASS are empty, python dashboard.py opens that same in-memory data with seven sample dogs. An instructor does not need a cluster to see the filter.

Security

Outcome 5 is the habit of treating a record as something you protect on purpose (Southern New Hampshire University [SNHU], n.d.). Contact exits if CONTACT_STAFF_PASSWORD and CONTACT_ADMIN_PASSWORD are missing. There is no default password in that source. A

missing token is 401 with {}. A staff token on a write is 403 with {}. The test fails if that body contains the sample phone number. AuditLog takes actor, action, id, and status. It does not take a phone or an address. AnimalShelter fails closed the same way. No client and no MONGO_USER or MONGO_PASS means it does not connect. An empty lookup on delete returns 0. The original CS-340 delete used many=True, which would wipe aac.animals. The original constructor on main still has those defaults. That file is the original submission. The enhanced module does not copy it.

How the artifacts fit together

The program asked for one artifact in software design, one in algorithms and data structures, and one in databases (SNHU, n.d.). I used a past project for each so the enhancement is visible against the original (Hopewell, 2026).

CS-320 is the shape of a service: rules, storage, HTTP, and a login. CS-300 is the choice of structure and the cost of that choice, including a schedule that has to respect prerequisites. CS-340 is where those ideas land in stored data: a collection, an index, a role, and a screen a staff member can open. Outcome 1 is the code review and the public branches. Outcome 2 is the narratives and this introduction. Outcome 3 is the HashMap versus SQLite choice and the vector versus BST versus hash timings. Outcome 4 is JUnit, SQLite, C++, PyMongo, pytest, and the HTTP layer. Outcome 5 is the auth filter, the audit log, and the AnimalShelter checks above.

The databases enhancement is the Milestone Four branch, including the demo mode so the dashboard runs from the zip. The 3-2, 4-2, and 5-2 grades did not ask for a second pass, so that branch is the databases work. Leftovers I would still name are the unique index against a dirty AAC dump, aggregation grouped on outcome_type, a dashboard that shows a table and a pie

chart (a map was in the early plan and is not in this build), and an insert payload that is not allow-listed. When credentials are set, live Mongo defaults to localhost port 27017.

Where the artifacts are

Software design, original: https://github.com/AmauriHopewell/CS320. Enhanced: https://github.com/AmauriHopewell/CS320/tree/cs499-milestone-two/enhanced. From that enhanced folder, mvn test is 37 tests. Narrative: CS-499_3-2_Narrative_Software_Design_Amauri_Hopewell.

Algorithms, original: https://github.com/AmauriHopewell/CS300. Enhanced: https://github.com/AmauriHopewell/CS300/tree/cs499-milestone-three/enhanced. Build with /O2. The enhanced README has the compare command. Narrative: CS-499_4-2_Narrative_Algorithms_Amauri_Hopewell.

Databases, original: https://github.com/AmauriHopewell/CS340. Enhanced: https://github.com/AmauriHopewell/CS340/tree/cs499-milestone-four/enhanced (commit beefb7b). From that folder: pip install -r requirements.txt, then python -m pytest tests -q, then python dashboard.py. No live Mongo is required for that path. Narrative: CS-499_5-2_Narrative_Databases_Amauri_Hopewell.

The code review video is the Module Two submission, CS499 2-2 Code Review Amauri Hopewell. It is posted at https://youtu.be/ZKbE1ZThCFY.

References

Hopewell, A. (2026, September 6). CS-499 Module One assignment: ePortfolio enhancement plan [Unpublished manuscript]. Southern New Hampshire University.

Southern New Hampshire University. (n.d.). CS 499 final project guidelines and rubric.

Acknowledgment of generative AI use

I used Grok (xAI) in October 2026 to help organize this self-assessment against the Module Seven prompt and to format the Word file. The artifacts, test counts, branch names, and job target are from my Module One plan and from the milestone work I submitted this term. I checked the three GitHub branch URLs before I kept them. I revised the status sentences on October 18, 2026, after the milestone grades came back without a request for more code.